MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

GUIDELINE FOR TREATMENT OF BACTERIAL MENINGITIS IN ADULTS

<table>
<tr><th colspan="3">Community-Acquired Meningitis</th></tr>
<tr><th>Common Pathogens</th><th>Empiric Treatment Regimen</th><th>Comments</th></tr>
<tr><td>N. meningitidis
S. pneumoniae
H. influenzae
(Herpes Simplex Virus^)</td><td><u>1st line:</u>
Ceftriaxone 2 g IV q12h
+ Vancomycin IV**
(+ Acyclovir 10mg/kg q8 hours^)
+ Dexamethasone 10 mg IV q6h

<u>Cephalosporin allergy:</u>
Meropenem* 2g IV q8h
+ Vancomycin IV**
(+ Acyclovir 10mg/kg q8 hours^)
+ Dexamethasone 10mg IV q6h</td><td rowspan="2">• ^While HSV is more commonly a cause of encephalitis rather than true meningitis, it can be challenging to distinguish these syndromes clinically. As HSV encephalitis is a common and serious but treatable cause of CNS infection, empiric acyclovir should be strongly considered in patients with concern for community-acquired CNS infection.

• CT prior to lumbar puncture if:
○ Immunocompromised sed
○ History of CNS disease (mass lesion, stroke)
○ New onset seizures
○ Papilledema
○ Abnormal level of consciousness
○ Focal neurologic deficit

• Dexamethasone should be administered 10-20 min <u>before</u> antimicrobial therapy for maximal efficacy. Benefit seen mainly in S. pneumoniae per observational data, can consider discontinuing for other etiologies.

• Infectious Diseases consultation is strongly recommended for patients with suspected meningitis or encephalitis

• Typical Duration of Therapy
○ N. meningitidis = 7 days
○ H. influenzae = 7 days
○ S. pneumoniae = 10-14 days
○ L. monocytogenes = <u>></u> 21 days
○ Aerobic GNRs = 21 days
○ HSV = 14-21 days

*Adjust dose based on renal function, see renal dose policy

**(see nomogram)</td></tr>
<tr><td>Listeria risk factors:
Pregnant or
<u>></u>50 y/r or
Immunocompromised

Ampicillin, meropenem, and TMP-SMX are all agents with coverage against listeria.</td><td><u>1st line</u>
Ampicillin* 2g IV q4h
+ Ceftriaxone 2g IV q12h
+ Vancomycin IV**
(+ Acyclovir 10mg/kg q8 hours^)
+ Dexamethasone 10mg IV q6h

<u>Penicillin or cephalosporin allergy:</u>
Meropenem*2g IV q8h
+ Vancomycin IV**
(+ Acyclovir 10mg/kg q8 hours^)
+ Dexamethasone 10mg IV q6h

OR

TMP-SMX* 5 mg/kg IV q8h
+ Ceftriaxone 2g IV q12h
+ Vancomycin IV**
(+ Acyclovir 10mg/kg q8 hours^)
+ Dexamethasone 10mg IV q6h

Note: TMP-SMX may not be indicated in pregnancy. Consult ID for recommendations</td></tr>
</table>

MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Healthcare-Associated Meningitis and Ventriculitis		
Common Pathogens	**Empiric Treatment Regimen**	**Comments**
Basilar skull fracture *S. pneumoniae* *H. influenza* Group A strep	1st line: **Ceftriaxone** 2g IV q12h **+ Vancomycin** IV** Cephalosporin allergy: **Meropenem*** 2g IV q8h + **Vancomycin** IV**	• Typical Duration of Therapy ○ CoNS/C. *acnes* with no or minimal CSF WBC, normal CSF glucose, and few clinical symptoms or systemic features: 10 days ○ CoNS/C. *acnes* with derangements in CSF WBC, glucose, or clinical symptoms or systemic features: 10-14 days ○ *S. aureus*/gram-negative bacilli: 14 days, may consider longer duration (21 days) ○ In all cases, if repeated positive CSF cultures on appropriate therapy, treatment should continue 10-14 days after last positive culture. • CSF shunt infections: Routine standard for infection clearance is removal of shunt. Prior to replacement of shunt, cultures should be negative for: ○ CoNS + normal CSF findings: 3days ○ CoNS + abnormal CSFfindings: 7 days ○ *S. aureus*: 10 days ○ Gram negative bacilli: 10-14days (plus) • Intraventricular therapy may be considered for patients in which the infection responds poorly to systemic therapy or unable to remove infected device. Please use "Intrathecal/Intraventricular Antibiotics" MiChart Order Set (amikacin, gentamicin, tobramycin, colistin, vancomycin, and amphotericin B deoxycholate orders pre-built) *Adjust dose based on renal function, see renal dose policy **(see nomogram)
Penetrating trauma *S. aureus* Coag-negative *Staphlococci* Aerobic gram-negative bacilli (e.g. *Psuedomonas*)	1st line: **Cefepime*** 2g IV q8h **+ Vancomycin** IV** Cephalosporin allergy: **Meropenem*** 2g IV q8h + **Vancomycin** IV**	
Post neurosurgery **And/or** **Presence of CSF shunt or drain (e.g. EVD device, VP shunt)** *S. aureus* Aerobic gram-negative bacilli (e.g *Psuedomonas*) Coag-negative *Staphylococci* *C. acnes*	1st line: **Cefepime*** 2g IV q8h **+ Vancomycin** IV** 2nd line/Alternative regimen to vancomycin: **Cefepime*** 2g IV q8h **+ Linezolid** 600mg IV q12h Cephalosporin allergy: **Meropenem*** 2g IV q8h + **Vancomycin** IV** or **Linezolid** 600mg IV q12h	

MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Antimicrobial Subcommittee Approval: 08/12/2025	Originated: Unknown
P&T Approval: 01/2026	Last Revised: 07/2025

Revision History:

03/21: Updated vancomycin dosing & hyperlinks

02/24: Updated allergy guidance; removed aztreonam alternative options; added linezolid as alternative to vancomycin for CSF shunt/neurosurgery infections

07/25: Updated recommendation for consideration of empiric HSV coverage

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source

If obtained from a source other than med.umich.edu/asp, please visit the webpage for the most up-to-date document..